RV Educational Institutions®

RV Institute of Technology and Management

(Affiliated to VTU, Belagavi)

JP Nagar 8^{th} Phase, Bengaluru - 560076

Department of Mechanical Engineering

RENEWABLE ENERGY POWER PLANTS (BME654B)

Course Name: RENEWABLE ENERGY POWER PLANTS

Course Code: BME654B

VI Semester

2025 Scheme

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VISION

Constantly endeavour to disseminate quality education in Mechanical Engineering with cutting edge technology leading to quantum leap in innovations for societal needs.

MISSION

- To produce quality mechanical engineers by providing ICT based state-of-the-art facilities for experiential learning.
- To bridge the gap between the Industry and Academia to make the students industry-ready by close interaction with Industry.
- To infuse best mechanical engineering practices through quality education, ingenuity, innovation and entrepreneurial skills leading to incubation ecosystem.
- To provide best academic experience to peruse their goals of higher education and research in top universities.

PROGRAM SPECIFIC OBJECTIVES (PSOS)

- PSO1: Apply the knowledge of material science, design, manufacturing technology and Thermal engineering to solve industry related problems.
- PSO2: Proficient to use all appropriate software to solve complex engineering problems.
- PSO3: Impart quality knowledge in the field of management and interdisciplinary technologies to suit industry and society.

Contents

Chapter 1

Tidal Power and Ocean Thermal Energy

Tidal Power: Tides and waves as energy suppliers and their mechanics; fundamental characteristics of tidal power, harnessing tidal energy, advantages and limitations.

Ocean Thermal Energy Conversion: Principle of working, OTEC power stations in the world, problems associated with OTEC.

1.1 Tidal Energy

1.2 Introduction

Tidal energy is produced by the surge of ocean waters during the rise and fall of tides. Tidal energy is a renewable source of energy harnessed by converting energy from tides into useful forms of power, mainly electricity using various methods.

1.3 Tides And Waves As Energy Suppliers And Their Mechanics

- Tidal power is taken from the Earth's oceanic tides. Tidal forces are periodic variations in gravitational attraction exerted by celestial bodies. These forces create corresponding motions or currents in the world's oceans.
- Due to the strong attraction to the oceans, a bulge in the water level is created, causing a temporary increase in sea level.

- As the Earth rotates, this bulge of ocean water meets the shallow water adjacent to the shoreline and creates a tide. This occurrence takes place in an unfailing manner, due to the consistent pattern of the moon's orbit around the earth.
- The magnitude and character of this motion reflects the changing positions of the Moon and Sun relative to the Earth, the effects of Earth's rotation, and local geography of the seafloor and coastlines.
- The rise of seawater is called high tide and fall in seawater is called low tide and this process of rising and receding of water waves happen twice a day and cause enormous movement of water. Thus, enormous rising and falling movement of water is called tidal energy, which is a large source of energy and can be harnessed in many coastal areas of the world.

1.4 Fundamental Characteristics of Tidal Power

- Tidal power or tidal energy is a form of hydropower that converts the energy obtained from tides into useful forms of power, mainly electricity.
- Although not yet widely used, tidal energy has potential for future electricity generation.
- Tidal power is the only technology that draws on energy inherent in the orbital characteristics of the Earth–Moon system, and to a lesser extent in the Earth–Sun system.
- Greater tidal variation and higher tidal current velocities can dramatically increase the potential of a site for tidal electricity generation.
- Tidal power is also relatively prosperous at low speeds, in contrast to wind power. Water has one thousand times higher density than air and tidal turbines can generate electricity at speeds as low as 1m/s, or 2.2mph.
- Tidal energy is clean, renewable, and sustainable form of energy resource. It has no impact on climate because it does not produce any greenhouse gases.

1.5 Harnessing Tidal Energy

- **Tidal stream generator**: Tidal stream generators make use of the kinetic energy of moving water to power turbines, in a similar way to wind turbines that use the wind to power turbines. Some tidal generators can be built into the structures of existing bridges or are entirely submersed, thus avoiding concerns over the impact on the natural landscape.

- **Tidal barrage**: Tidal barrages make use of the potential energy in the difference in height (or hydraulic head) between high and low tides. When using tidal barrages to generate power, the potential energy from a tide is seized through the strategic placement of specialized dams. When the sea level rises and the tide begins to come in, the temporary increase in tidal power is channeled into a large basin behind the dam, holding a large amount of potential energy. With the receding tide, this energy is then converted into mechanical energy as the water is released through large turbines that create electrical power through the use of generators

- **Dynamic tidal power:** Dynamic tidal power (or DTP) is a theoretical technology that would exploit an interaction between potential and kinetic energies in tidal flows. It proposes that very long dams (for example: 30–50 km length) be built from coasts straight out into the sea or ocean, without enclosing an area

- **Tidal lagoon**: A new tidal energy design option is to construct circular retaining walls embedded with turbines that can capture the potential energy of tides. The created reservoirs are similar to those of tidal barrages, except that the location is artificial and does not contain a pre-existing ecosystem. The lagoons can also be in double (or triple) format without pumping or with pumping that will flatten out the power output .

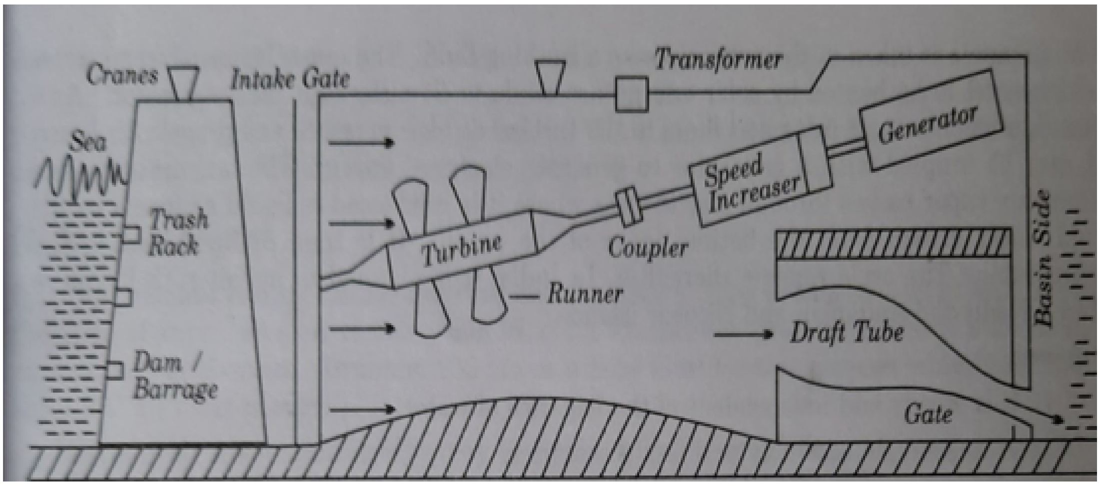

Fig. 1.1 Schematic layout of tidal power house

The above layout (1.1) is the tidal power house or tidal power plant that consists of a barrage built across the tidal reach to create pool in which water can be stored. Inside the barrage reversible water turbines and flood gates are installed. During the occurrence of high tide and low tides, the flood gates open and close respectively.

One tide cycle is identified as the shift from low tide to high tide and back to low tide. The time duration to complete one tidal cycle is approximately 12.5 hours. During this period the pool is filled and empties. Theses may be classified as single basin system and double basin system

1.5.1 Limitations

- Economic recovery of energy from tides is feasible only at those sites where energy is concentrated in the form of tidal range of about 5 m or more and geography provide favorable site for economic construction of a tidal plant, thus it is site specific
- Due to mismatch of lunar driven period of 12.5 hours and human (solar) period of 24 hours, the optimum tidal power generation is not in phase with demand
- Changing tidal range in two weeks period produces changing power
- The turbines are required to operate at variable heads

- Requirement of large water volume flow at low head necessitates parallel operation of many turbines
- Tidal plant disrupts marine life at the location and can cause potential harm to ecology
- It requires very large capital cost at most potential installations
- The location of sites may be distant from the demand centers

1.6 Ocean Thermal Energy

About 70% of earth's surface is covered by ocean which is continuously heated by solar heat. Solar heat is stored as uneven distribution of heat between warm surface water and cold deep ocean water (called gradient) from where it is harnessed as ocean thermal energy

1.7 Principle Of Working

The basic principle of ocean thermal energy conversion (OTEC) is explained as follows:

Closed cycle OTEC

- The warm water from the ocean surface is collected and pumped through the heat exchanger to heat and vaporize a working fluid, and it develops pressure in a secondary cycle.
- Then, the vaporized working fluid expands through a heat engine (similar to a turbine) coupled to an electric generator that generates electrical power.
- Working fluid vapor coming out of heat engine is condensed back into liquid by a condenser.
- Cold deep ocean water is pumped through condenser where the vapor is cooled and returns to liquid state.
- The liquid (working fluid) is pumped again through heat exchanger and cycle repeats. It is known as closed-cycle OTEC.

Open cycle OTEC

- If ocean surface water is high, enough propane or similar material is used as working fluid; otherwise, for low-temperature surface water, fluid such as ammonia with low boiling point is used.
- In an open-cycle OETC, warm ocean surface water is pumped into a low-pressure boiler to boil and produce steam.

- Then, the steam is used in steam turbine to drive an electrical generator for producing electrical power. The cold deep sea water is used in condenser to condense steam.
- Some fractions of electrical power generated by OTEC plants are used for operating and controlling equipment involved in power plants, and high electrical power is used for feeding to several other energy consumers.

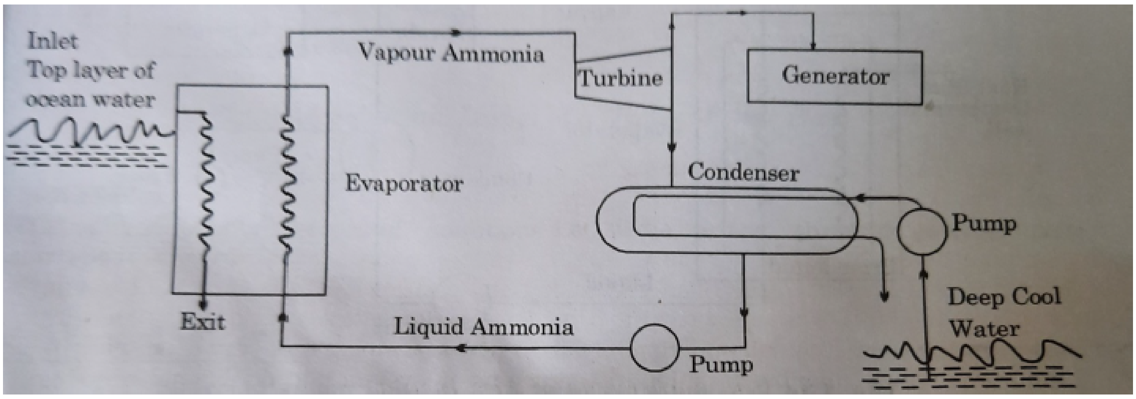

Fig. 1.2 Ocean thermal energy conversion – open cycle system

The above unit represents the open cycle type of ocean thermal energy conversion system that consists of an evaporator, a turbine, a generator, a condenser and two pumps.

A low boiling point liquid i.e. ammonia is taken into the evaporator as a working fluid. The upper layers of ocean water which need to be heated by solar energy are made to flow through the evaporator. As a result, ammonia evaporates and flows to the turbine at high pressure and propels it.

Later it may be coupled with a generator to generate electrical energy.

The low pressure exit ammonia vapor passes through the condenser where it is condensed to liquid ammonia by the cold water drawn from the bottom layer of the ocean. It is then pumped back to the evaporator. The cycle repeats thereafter.

1.8 Rankine Cycle

The basic Rankine cycle shown in the below Figure that consists of an evaporator, a turbine expander, a condenser, a pump, a working fluid.

- In open-cycle OTEC, warm sea water is used as working fluid, whereas in closed-cycle type, low- boiling point ammonia or propane is

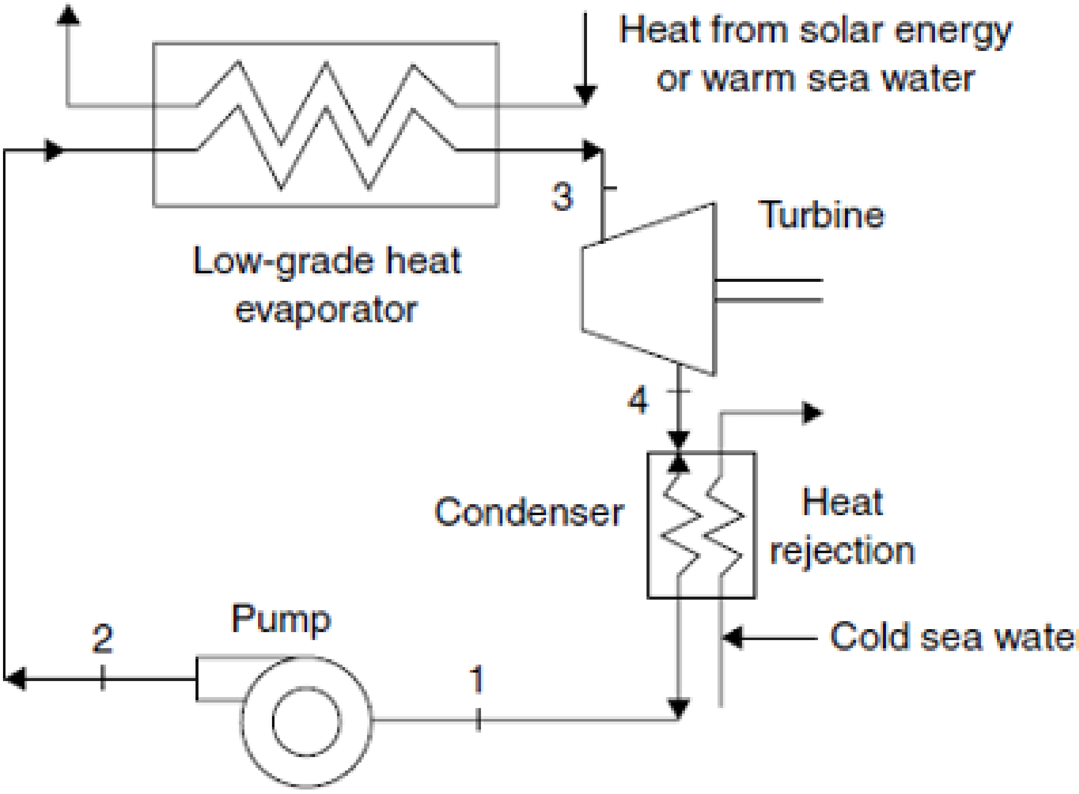

Fig. 1.3 OTEC Rankine Cycle

- Warm ocean surface water flows into the evaporator which is the high-temperature heat source. A fluid pump is utilized to force the fluid in a heat evaporator where liquid fluid vaporizes.
- Then, the vapor of boiling fluid enters the turbine expander coupled with an electrical generator to generate electrical power.
- The vapor released from the turbine enters into condenser where it condenses. The cold deep sea water is pumped through the condenser for heat rejection from vapor fluid and condenses it as liquid fluid.
- The liquid fluid is again pumped through evaporator and cycle repeats. As temperature difference between high- and low-temperature ends is large enough, the cycle will continue to operate and generate power.

1.9 OTEC Power Stations In The World

- Makai Ocean Engineering's ocean thermal energy conversion (OTEC) power plant in the US is the world's biggest operational facility of its kind with an annual power generation capacity of 100kW, which is sufficient to power 120 homes in Hawaii.
- OTEC plant in Japan overseen by Saga University

1.10 Problems Associated With OTEC

- **High cost**: Electricity generated by OTEC plants is more expensive than electricity produced by chemical and nuclear fuels.
- **Complexity**: OTEC plants must be located where a difference of about 20°C occurs year round. Ocean depths must be available fairly close to shore-based facilities for economic operation. Floating plant ships could provide more flexibility.
- **Acceptability:** For the large-scale production of electricity and other products, OTEC plants are poorly acceptable due to their high costs.
- **Ecosystem damage**: It is obvious by setting OTEC plants.
- **Lower efficiency**: A higher temperature difference between ocean surface warm water and cold deep ocean water is required for highly efficient operation of plant.